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Review

Adaptive mesh refinement for two-phase slug flows with an *a priori* indicator

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ABSTRACT

The purpose of this paper is to present a novel adaptive mesh refinement *AMR* technique for computing unstable one dimensional two-phase flows in pipelines. In multiphase flows, the prediction and localisation of inter-facial waves, slugs and instabilities related to flow conditions under study require high levels of accuracy. This is more apparent in systems at industrial scales, where flow lines possess highly distorted regions and irregular topologies.

Uniform fine meshes for these long devices are costly and in general situations the optimum space discretisation could not be determined *a priori*.

Adaptive mesh refinement *AMR* procedure provides a remedy to this problem by refining the mesh locally, allowing to capture regions where sharp discontinuities and steep gradients are present. With appropriate algorithm and data organisation, *AMR* helps to reduce CPU time and speeds up simulations of flows in long pipes. The effectiveness of *AMR* methods relies on estimators that determine where refinement is required. We show in this work that for transient flows combining gradient-based error estimator with *Kelvin–Helmholtz* stability condition can improve the acceleration of computation and locate regions where refinements are required. The *Kelvin–Helmholtz* is a local condition and is an *a priori* indicator for the refinement.

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1. Introduction

The difficulties of modelling multiphase flows in long pipes reside in the capturing profiles of interfaces and discontinuities associated with flows in extended systems that can exceed several kilometres. These irregularities result from flow conditions combined with the nature of the topology of the terrain. Slugs and waves that characterise such systems induce sudden mixing of phases and abrupt accelerations and decelerations. Simulations of these problems and their complexities are approached in one space dimension for industrial scales. For modelling industrial multiphase flows, various formulations use multi-fluid equations. These formulations are based on multi-field transport equations and their closure relationships (constitutive laws) rely on algebraic relations.

In this work the model considered is the derived version developed and detailed in Drew and Passman (1998) where the macroscopic balance laws were reduced to local partial differential equations by averaging techniques. For details, see for example Vernier and Delhaye (1968), Ishii (1975), Delahaye and Achard (1977), Nigmatulin (1979), Banerjee and Chan (1980).

The adaptive mesh refinement *AMR* technique is now a mature approach since it was pioneered by Pereyra and Sewell (1975), Oliger (1979), Berger and Oliger (1984), Berger and Collela (1989), Quirk (1991), Sussman et al. (1999) and many others.

One of the crucial parts of any *AMR* method is the availability of an efficient and consistent error estimator to predict accurately the error in the solution.

Several heuristic error estimation approaches have been proposed by Babuska and Rheinboldt (1978), Zienkiewicz and Zhu (1987), Oden et al. (1989) more recently Oden et al. (2003) and Prudhomme et al. (1989) are amongst many other researchers. Most of the functional error estimates are *a posteriori*, *a priori* error estimates are usable mainly for theoretical studies, this is due to the fact that the information available on the solution before the process is insufficient or often is not available. An efficient error indicator has to be computationally economical, simple to implement and finally insensitive to problem under study.

Adaptive mesh refinement procedure is performed in order to keep execution time within acceptable interval and a reasonable accuracy concerning the prediction of the flow distribution and behaviour along the system.

Modelling flow regimes (Stratified, Slug, Bubbly, etc.) in horizontal or vertical pipes lead to various conservation laws that are differentiated with various source terms. The nature and the character of these source terms are major factors in the loss of hyperbolicity and hence dictating the choice of adequate numerical scheme to describe appropriately the physics in the model, Leveque (1998) and Dumbser et al. (2008). In the last three decades, new classes of upwind schemes that are robust and stable have been developed, these are named Advection Upstream Splitting Method *AUSM* initiated by Liou and Steffen (1991) and Wada and Liou (1994). Later, *AUSM*-familly schemes were introduced in Liou (1996) and Liou (2006) and tested in different situations in Tiselj and Petelin (1997), Evje and Fjelde (2003), Evje and Flåtten (2003) amongst many other researchers.

In this work, we consider isothermal two fluids model to simulate the slug flow regime in long pipes with bumpy topology. The resulting system of governing equations is solved with a version of *AUSMDV** scheme that is according to Evje and Flåtten (2003) robust and suitable for slow transitions occurring in multiphase flows. The *AMR* strategy used here is based on the gradient of solutions and the *Kelvin–Helmholtz* stability condition derived by Barnea and Taitel (1993). This latter condition allows for refinement in localised regions. The originality of the present approach resides in its simplicity to implement and furthermore in the use of an *a priori* indicator concerning slug detections.

2. Two-fluid model

2.1. Governing equations

Expressions of the ensemble-averaged continuity and momentum equations for isothermal two-phase flow with ideal interface (no mass transfers, no chemical reactions, no phase change) are given below

$$\begin{cases} \frac{\partial}{\partial t}(\alpha_k\bar{\rho}_k)+\nabla\cdot(\alpha_k\bar{\rho}_k\bar{u}_k)=0 \\ \frac{\partial}{\partial t}(\alpha_k\bar{\rho}_k\bar{u}_k)+\nabla\cdot(\alpha_k\bar{\rho}_k\bar{u}_k\bar{u}_k+\alpha_k\overline{P}_k)=P_{ki}\cdot\nabla\alpha_k+\nabla\cdot\alpha_k(\sigma_k+\bar{\tau}_k)+\alpha_k\bar{\rho}_k g_k+\psi_{ki} \end{cases} \tag{1}$$

Above, k is the index of the phase, α_k is its volume fraction while $\bar{\rho}_k, \bar{u}_k$ represent its density and velocity, $\overline{P}_k$ refers to the pressure, $\bar{\tau}_k$ represents stress that is different from Reynolds turbulent stress. All these quantities are estimated via a judicious averaging process that involve time or space or ensemble, see Drew and Passman (1998). σ_k represents the viscous stress tensor. $\alpha_k\bar{\rho}_k g_k$ is the body force, P_{ki} contains inter-facial pressure and ψ_{ki} represents inter-facial and boundary momentum rates. Here, the over-bar symbol refers to phasic or weighted averaged quantity. We recall that for any physical quantity Ω, its weighted averaged (phasic) counterpart is given by

$$\overline{\Omega}_k = \langle X_k\cdot\Omega\rangle/\alpha_k$$

where

$$\langle X_k\cdot\Omega(x,t)\rangle = \frac{1}{T}\int_t^{t+T} X_k\cdot\Omega(x,\tau)\mathrm{d}\tau \tag{2}$$

$$\alpha_k = \langle X_k(x,t)\rangle$$

Above, T is the sampling time while the characteristic distribution of the phase k is defined as

$$X_k(x,t)=\begin{cases}1 & \text{if } x \text{ is in phase } k \text{ at time} t\\ 0 & \text{otherwise}\end{cases}$$

2.2. Model formulation

For numerical modelling, assumptions and simplifications are needed: we consider one dimensional flow of two phase gas and liquid. Dropping the weighted-average notation, conservations of immiscible phases reduce to

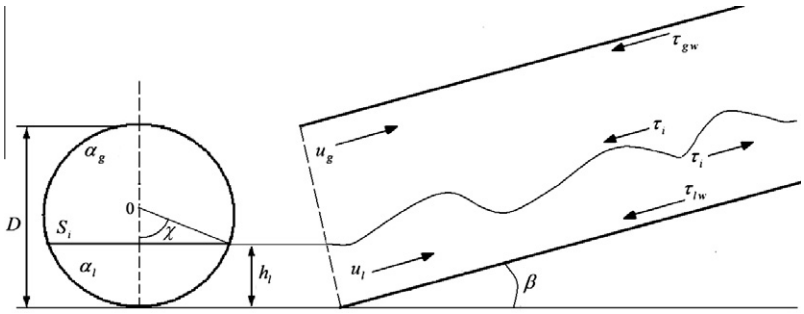


Fig. 1. Schematic view representing two phase flow in a pipe.

$$\begin{cases} \frac{\partial}{\partial t}(\alpha_g \rho_g) + \frac{\partial}{\partial x}(\alpha_g \rho_g u_g) = 0 \\ \frac{\partial}{\partial t}(\alpha_l \rho_l) + \frac{\partial}{\partial x}(\alpha_l \rho_l u_l) = 0 \end{cases} \tag{3}$$

We consider inviscid phases; $\sigma_k = 0$ and non-turbulent flows; $\bar{\tau}_k = 0$.

The bulk-phase pressure $\overline{P}_k$, the pressure jump between phases and the interface is included in the non-conservative term; $P_{ki} \cdot \frac{\partial \alpha_k}{\partial x}$. Modelling and interpreting pressure forces arising from averaged two-phase flow equations is difficult. Using averaging theorem, Prosperetti and Jones (1984) showed that P_{ki} is a consequence of boundary integral along the interfaces. In Sha and Soo (1979) and in Stuhmiller (1977) investigations on the influence of the inter-facial pressure forces on the character of the equations of two-phase flow models and their influence were outlined. Several expressions of this term exist in the literature, see for example, Soo (1990), Bestion (1990) and Saurel and Abgrall (1999). We consider the derivation given in Taitel and Dukler (1977), Gu and Guo (2007). For a system inclined to the horizontal line with an angle β, that component is expressed by

$$\begin{aligned} P_{ki} \cdot \frac{\partial \alpha_k}{\partial x} &= \left[P_k - \epsilon_k \cdot \rho_k g \cos\beta \cdot \left(\alpha_k \cdot \frac{\partial h_k}{\partial \alpha_k} \right) \right] \cdot \frac{\partial \alpha_k}{\partial x} \\ &= [P_k - \epsilon_k \cdot \rho_k g \cos\beta \cdot D_k] \cdot \frac{\partial \alpha_k}{\partial x} \end{aligned} \tag{4}$$

Resulting in the momentum exchange interaction due to interface phase height or gas void gradients $\frac{\partial \alpha_k}{\partial x}$. The term $D_k = \alpha_k \frac{\partial h_k}{\partial \alpha_k}$ signifies the hydraulic depth of the phase k, h_k is the phase hold up while $\epsilon_k = \pm 1$ for each phase. A schematic a view of two-phase flow in an inclined pipe and phase distributions across pipe section are depicted in Fig. 1.

Normalised hydraulic depths $\widetilde{D}_k = D_k/D$ with respect to the pipe diameter D are expressed via geometric relationship relating the volume fraction and the liquid height to the wetted angle χ as follow

$$\begin{cases} \alpha_l = \frac{1}{\pi} \cdot (\chi - \frac{1}{2} \cdot \sin 2\chi) \\ \frac{1}{D} \cdot \frac{\partial h_l}{\partial \alpha_l} = \frac{\pi}{4 \cdot \sin\chi} = \widetilde{D}_l + \widetilde{D}_g \end{cases} \tag{5}$$

The profiles of $\widetilde{D}_k$ as function of wetted angle are represented in Fig. 2 below.

Concerning the last term ψ_{ki} in Eq. (1); let τ_i and τ_{kw} be the inter phase shear stress and wall shear stress for phase-k. S_i and S_{kw} representing the interface perimeter, and wetted perimeters per cross section area of the pipe respectively.

ψ_{ki} is decomposed into inter-phases and boundary interaction terms. Its components contain drag and friction forces, these are algebraic terms and are functions of Reynolds numbers and wetted perimeters, the nature of the flow of each phase (laminar/turbulent) is enclosed in these terms and is approximated as; $\psi_{ki} = \pm\tau_i \cdot S_i - \tau_{kw} \cdot S_{kw}$, it follows that momentum balances of gas and liquid phases simplify to

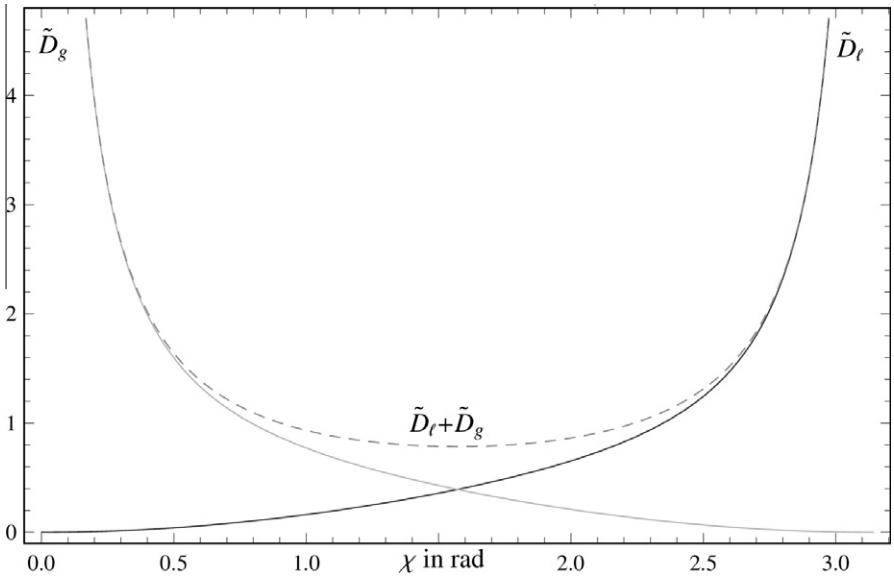


Fig. 2. Normalised hydraulic depths: Liquid (Black), gas (Grey) and $\widetilde{D}_l + \widetilde{D}_g$ (dashed).

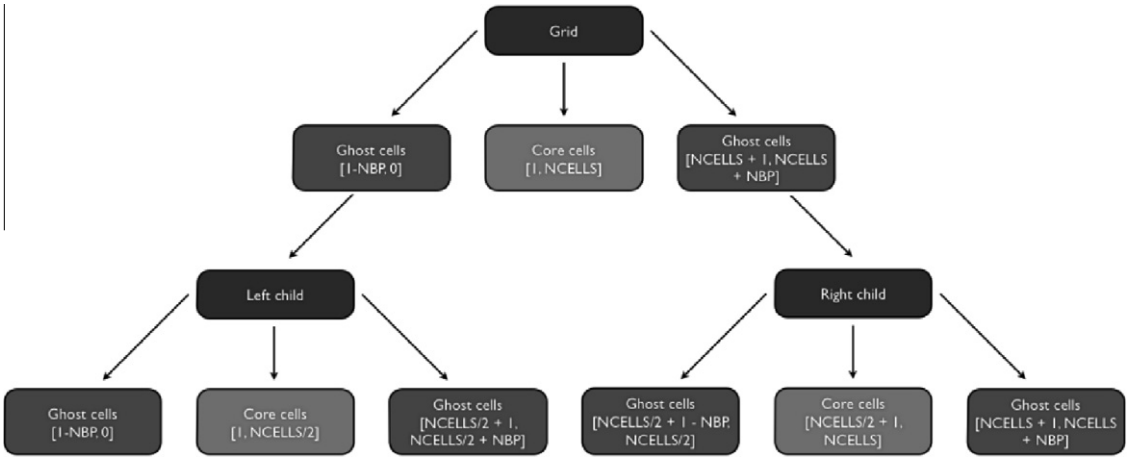


Fig. 3. Sketch of hierarchical *Parents*/*Childs* grids.

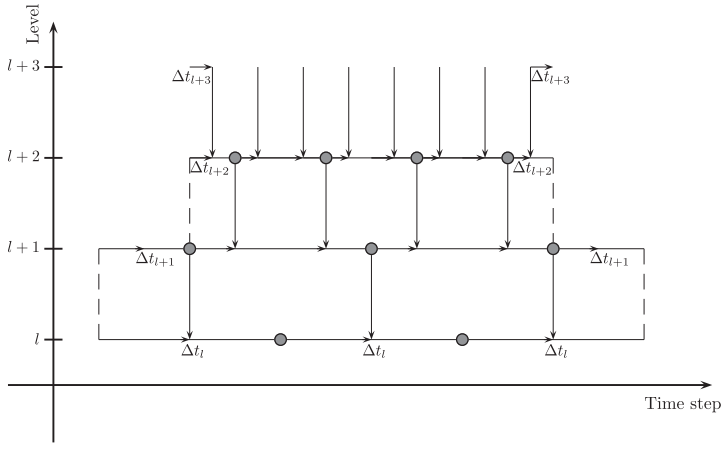


Fig. 4. Graph of four sequential *space*/*time* steps discretisation as: *level*/*time step* grids.

$$\begin{cases} \frac{\partial}{\partial t}(\alpha_g \rho_g \cdot u_g) + \frac{\partial}{\partial x}(\alpha_g \rho_g u_g^2 + \alpha_g P_g) = \left(P_g - \rho_g g \cos\beta \cdot D_g\right) \cdot \frac{\partial \alpha_g}{\partial x} + \tau_i \cdot S_i - \tau_{gw} \cdot S_{gw} - \rho_g \alpha_g g \sin\beta \\ \frac{\partial}{\partial t}(\alpha_l \rho_l \cdot u_l) + \frac{\partial}{\partial x}(\alpha_l \rho_l u_l^2 + \alpha_l P_l) = (P_l + \rho_l g \cos\beta \cdot D_l) \cdot \frac{\partial \alpha_l}{\partial x} - \tau_i \cdot S_i - \tau_{lw} \cdot S_{lw} - \rho_l \alpha_l g \sin\beta \end{cases} \tag{6}$$

Above ρ_k, α_k and u_k represent the density, the volume fraction and the velocity of the phase $k = g, l$, see Fig. 1. P_k stands for its pressure that can be an equation of state (EOS). For example, ideal gas, Tait–Murnaghan for water or a table EOS. Incompressible liquid and ideal gas are assumed here and single bulk pressure model is considered, $P_g = (\gamma - 1)\rho_g \cdot c_v T$.

2.3. Wall shear force

The wall shear force represents stresses acting on each phase at the wall. Different methodologies have been proposed to estimate it and have been reviewed by Simonič (1992). Wall shear stress is based on closure laws derived from fully developed steady state two-phase flows, and given by

$$T_{kw} = -\tau_k \cdot \frac{s_k}{A} = -\tau_k \cdot S_k \tag{7}$$

where A is the cross section area, $s_k = A \cdot S_k$ is the wetted wall perimeter by the phase k and τ_k is the wall shear stress on the phase. The constitutive relation for the wall shear stress is given in term of the standard wall friction factor f_k that was shown as:

$$\tau_k = \frac{1}{2} f_k \rho_k u_k |u_k| \tag{8}$$

Friction factors are modelled using single-phase pipe correlation that is function of Reynolds number.

3. Numerical solver

As mentioned in the introduction *AUSMDV** scheme is used, for detailed descriptions of the scheme, see Evje and Flåtten (2003).

If we let U be the vector of unknown fields, $F(U)$ signifies the flux vector, $H(U)$ the non-conservative coupling matrix and $S(U)$ be the inter-phase and wall frictions term, The set of balance laws in Eqs. (3) and (6) in compact form become

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} = H(P) \cdot \frac{\partial Q}{\partial x} + S(U, Q) \tag{9}$$

The construction of *AUSMDV** scheme starts by decomposing the flux vector $F(U) = F^c(U) + F^p(U)$ into convective and pressure flux component for each phase k as follow

$$F_k^c(U) = \begin{pmatrix} \alpha_k \rho_k \cdot u_k \\ \alpha_k \rho_k \cdot u_k^2 \end{pmatrix} \quad \text{and} \quad F_k^p(U) = \begin{pmatrix} 0 \\ \alpha_k \cdot P \end{pmatrix}$$

Table 1
Cases references and Air–Water physical properties.

Case reference	Air density (kg/m^3)	Water density (kg/m^3)	Temperature (K)	Pressure bar
22	1.25	998.4	292.5	1.034
36	1.23	998.2	293.5	1.024
37	1.02	998.4	292.5	1.020

Table 2
Initial conditions at the inlet: Air–water superficial velocities and liquid holdup.

Case reference	Mixture flow rate (m/s)	Air flow rate (m/s)	Water flow rate (m/s)	$\alpha_l^{equilbrium}$ in (%)
22	4.54	4.016	0.519	67.0
36	2.07	1.548	0.519	80.8
37	3.67	3.135	0.534	71.5

3.1. Source terms discretisation

The source terms on the rhs of the system (9) are discretised using the following second order spatial centred scheme.

$$\Sigma_k(U,Q,P)_j = H_k(P_j)\cdot\left(\frac{\widehat{Q}_j-\widehat{Q}_{j-1}}{\Delta x}\right)_k + S_k(U_j,Q_j) \tag{10}$$

where Δx is the cell size, $\widehat{Q}_j=(Q_j+Q_{j+1})/2$ with

$$Q_k=\begin{pmatrix}1\\ \alpha_k\end{pmatrix},\quad S_k(U,Q)=\begin{pmatrix}0\\ \pm\tau_i\cdot S_i-\tau_{kw}\cdot S_{kw}-\alpha_k\rho_k\cdot g sin\beta\end{pmatrix}$$

and the coupling sub-matrix

$$H_k(P)=\begin{pmatrix}0 & 0\\ 0 & P_k-\epsilon_k\cdot\rho_k\cdot g\cos\beta\cdot D_k\end{pmatrix}$$

Note that when an EOS is available for the phase k, $P(\rho_k, T)$ the density ρ_k can be substituted in $H_k(P)$ and in $S_k(U, Q)$.

Collecting different discretised components in (9), the form of advancing explicitly the solution in time internals $t\in[t_n, t_{n+1}]$ with a time step Δt

$$U_{kj}^{n+1}=U_{kj}^{n}-\frac{\Delta t}{\Delta x}\cdot\left[\left(F_{j+1/2}^{c}-F_{j-1/2}^{c}\right)+\left(F_{j+1/2}^{p}-F_{j-1/2}^{p}\right)\right]_k+\Delta t\cdot\Sigma_k(U_j,Q_j,P_j) \tag{11}$$

4. Adaptive mesh refinement *AMR*

AMR technique refines the numerical solution of the systems (11) in space and time on the initial grid $\mathcal{G}(\Omega)$ of the domain Ω. The technique used here was initiated by Berger and Collela (1989), Quirk (1991) and is based on hierarchy of grids $\mathcal{G}_l$ ranging from coarsest, labelled level zero, $l_0=0$ to finest, labelled $l=l_{max}$. Below, Fig. 3 depicts a schematic sequences of *Parents*/*Childs* meshing of the domain.

Each of these grids $\mathcal{G}_l$ contains patches $\mathcal{G}_{l,m}$ with $1\leqslant m\leqslant np$ that are *genuinely nested* by mean that the union of grids at level l is contained in the union of grids at level $(l-1)$ for $0\leqslant l\leqslant l_{max}$. In other words, a grid at level l is always nested within a grid at $(l-1)$ in a way that there is a buffer region of level $(l-1)$ cells surrounding the cells of level l at boundaries.

A grid entity at level l is formed of the union of single or several patches m at the same level, the formal description of hierarchical grids obeys to

$$\begin{aligned}&\mathcal{G}=\bigcup_l\mathcal{G}_l\\ &\mathcal{G}_l\subseteq\mathcal{G}_{l-1}\quad\text{and}\\ &\mathcal{G}_l=\bigcup_{m=1}^{m=np}\mathcal{G}_{l,m}\end{aligned} \tag{12}$$

Each level has a given resolution. The refinement ratio between successive levels, is constant and even; $r=\frac{\Delta x_{l+1}}{\Delta x_l}=\ldots=\frac{\Delta x_1}{\Delta x_0}$.

The refinement factor in time for explicit schemes has to be calculated for each level according the stability and accuracy criteria. In order to fulfil the stability condition ($CFL<1$), the time step must

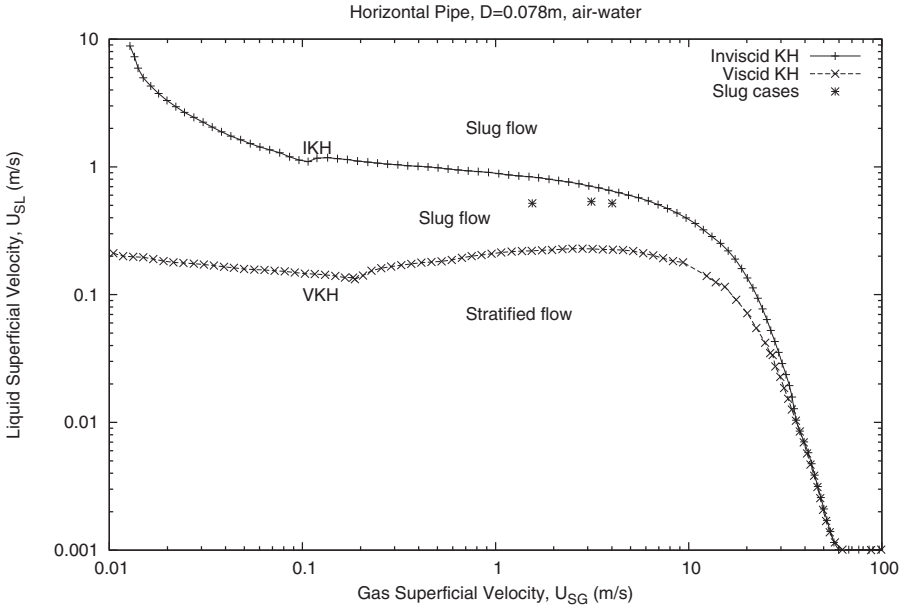


Fig. 5. Map of inviscid/viscous Kelvin–Helmholtz transitions from stratified to slug flow, from Barnea (1991).

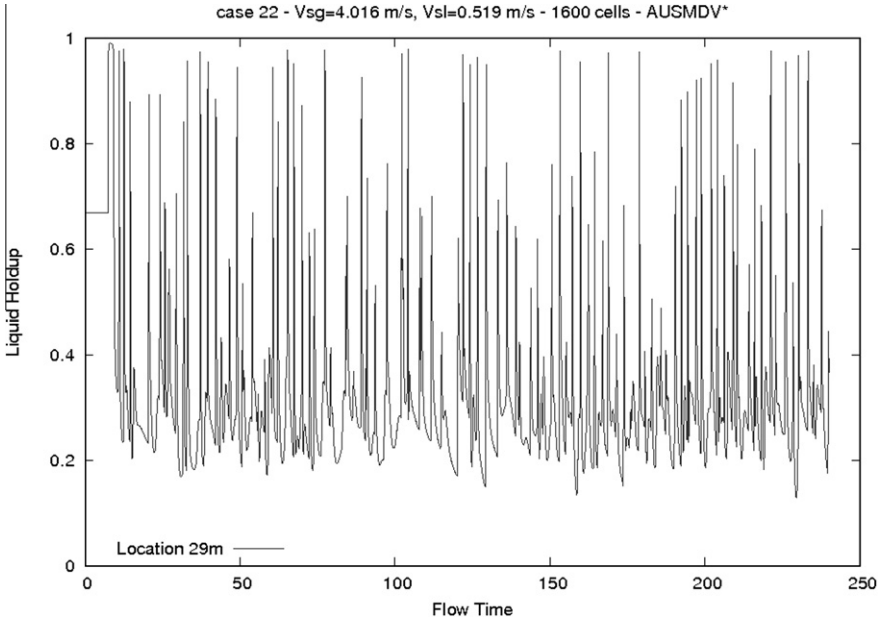


Fig. 6. Liquid holdup profile monitored at 29 m from inlet (1600 uniform cells).

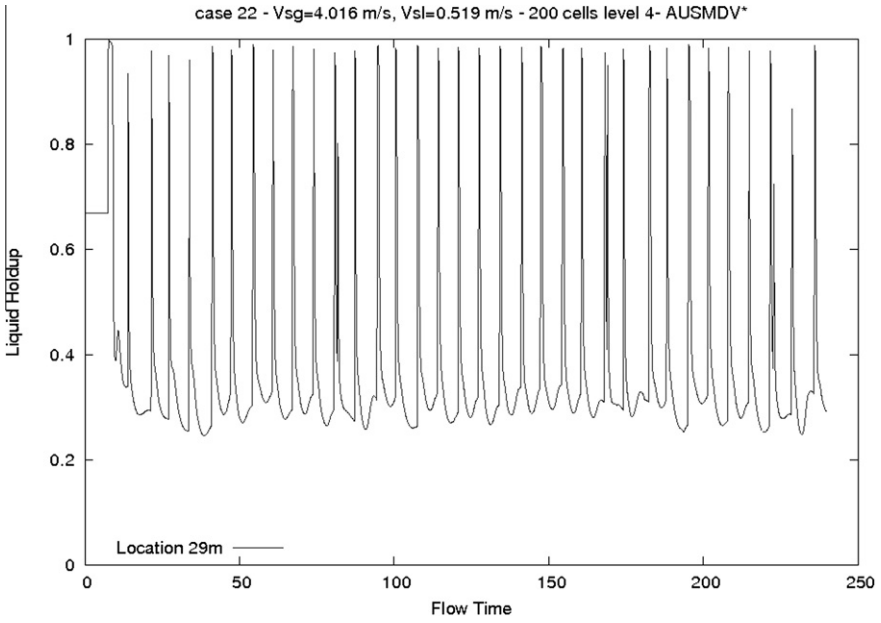


Fig. 7. Time trace of liquid holdup at 29 m (Level 4 based on 200 cells, with only gradient error criteria is applied in *AMR*).

be cycled, thus $r = \frac{\Delta t_{l+1}}{\Delta t_l}$. Fig. 4 shows successive steps of process leading to adaptive meshing in space and time step adjustment with a refinement ratio $r = 2$.

4.1. Refinement criterion in AMR

Diverse strategies of the refinement criteria for *AMR* technique based on local errors estimate can be applied in order to process mesh refinement. Most of these strategies are *a posteriori* type. At each time step and at every level, cells with error exceeding a prescribed threshold are flagged for refinement. The choice of an *optimal* criteria that possess reliability and effectiveness, see Carstensen (2004) is by no mean straightforward since *AMR* always implies an adequate balance between accuracy and efficiency.

Most of the error estimators rely on the truncation errors of the solution between different levels Berger and Oliger (1984) and Berger and Collela (1989), some approaches are based on monitoring local gradients other use second derivative of the solution as in *FLASH* package, see Fryxell et al. (1991) or a combination of the two as in *NIRVANA* code, see Ziegler (2008).

For slug flows in long pipes, monitoring gradients of the solution in our case is suitable to follow moving structures of liquids and it is desirable to furthermore consider the temporal change of these gradients.

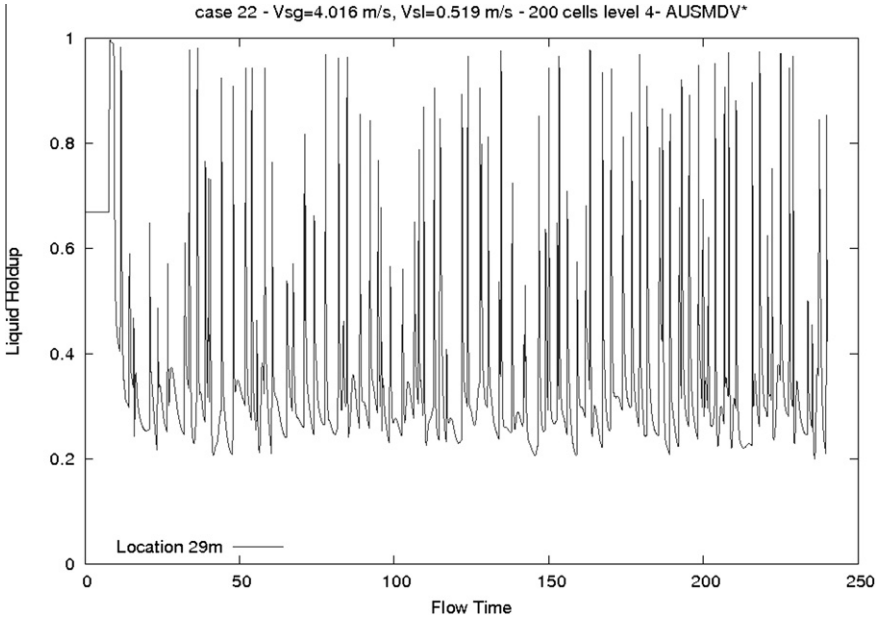


Fig. 8. Time trace of liquid holdup at 29 m (Level 4 based on 200 cells, the complete criteria is applied in AMR).

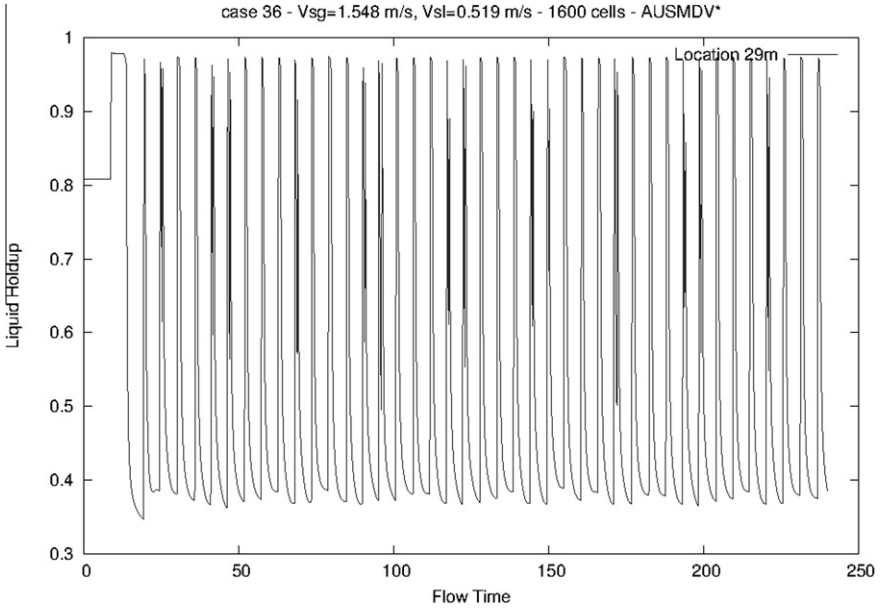


Fig. 9. Time trace of liquid holdup at 29 m (uniform mesh 1600 cells).

4.1.1. Gradient error and indicator

It is expected that the gradient of the field or flux will be steepest at regions where the rate of variation of the solution is significant compared to other regions. The gradient of the error can be used as criterion for the local as well as for the global error estimation. This latter is computed using the cell-wise local error monitor. An example of an *a posteriori* global error gradient detector is defined in the following way.

Let U_j, U be the numerical and the exact solution of the field in the cell j respectively. The true error in the gradient is $e_j = |\nabla U_j - \nabla U|$. Using the L_2 norm, the cell-wise gradient error (local) and the global error norm of the gradient are expressed respectively by

$$\begin{aligned} \|e_j\|_{L_2} &= \left\{ \int_{\Omega_j} e^T \cdot e \cdot d\Omega \right\}^{\frac{1}{2}} = \eta_j \\ \|e_\Omega\|_{L_2} &= \left\{ \sum_{j=1}^{N} \|e_j\|_{L_2}^2 \right\}^{\frac{1}{2}} = \eta_\Omega \end{aligned} \tag{13}$$

The exact solution U and its gradient are unknown and impossible to compute. Estimations in (13) are useful in theoretical investigations, in practice a *pseudo – solution* has to be recovered by some reconstruction method.

4.1.2. The Zienkiewicz–Zhu (ZZ) estimator

There are different approaches for rebuilding an error estimator. In this work, The *Zienkiewicz–Zhu*, η^{ZZ} estimator Zienkiewicz and

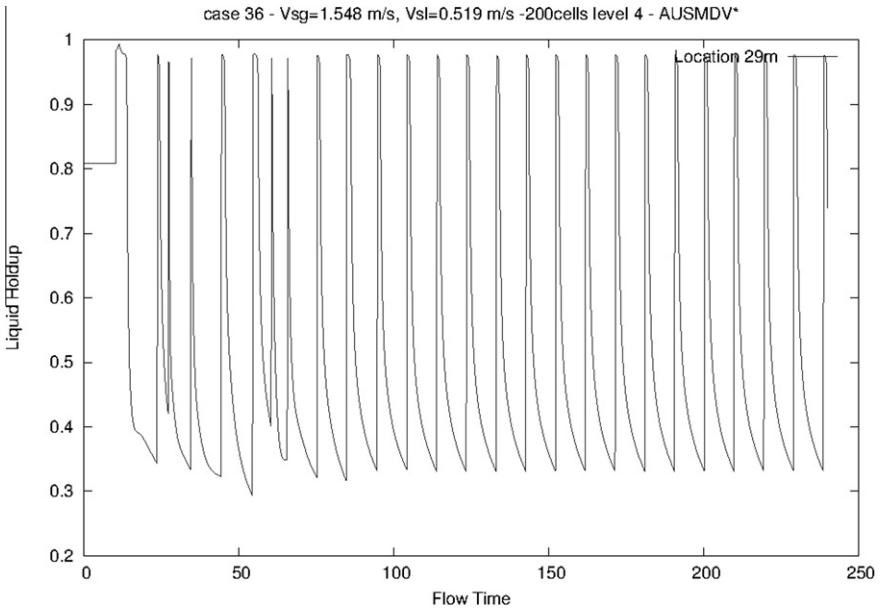


Fig. 10. Time trace of liquid holdup at 29 m, only gradient criteria is applied in *AMR*.

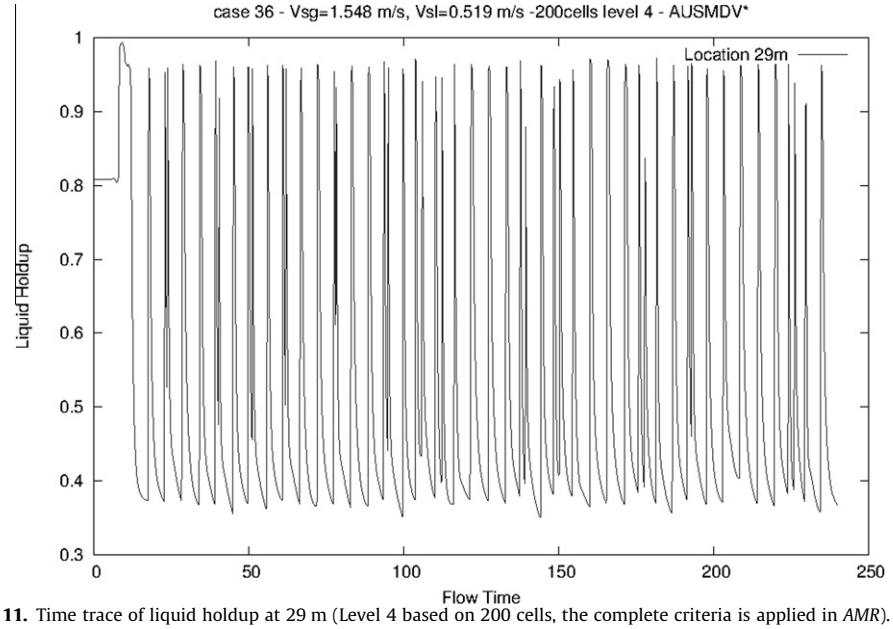


Fig. 11. Time trace of liquid holdup at 29 m (Level 4 based on 200 cells, the complete criteria is applied in *AMR*).

Zhu (1987) is briefly described, for rigorous and detailed analyses, readers are advised to consult the work in Ainsworth et al. (1992) and Zienkiewicz and Zhu (1992).

This error estimator is widely used because of its insensitivity to the problem at hand and is relatively simple to implement. Its aim is to build a recovery gradient GU of the gradient ∇U that is constructed based only on nodal values U_j, the absolute local error is $\tilde{e}_j = |GU_j - \nabla U|$. The local and global absolute error norms are then given by

$$\begin{aligned} \|\tilde{e}_j\|_{L_2} &= \left\{ \int_{\Omega_j} \tilde{e}^T \cdot \tilde{e} \cdot d\Omega \right\}^{\frac{1}{2}} = \eta_j^{ZZ} \\ \|\tilde{e}_\Omega\|_{L_2} &= \left\{ \sum_{j=1}^{N} \|\tilde{e}_j\|_{L_2}^2 \right\}^{\frac{1}{2}} = \eta_\Omega^{ZZ} \end{aligned} \tag{14}$$

Let $|\Omega_j|$ be the area of the cell and $|\omega| = \bigcup_{j\in\omega}|\Omega_j|$ is the union of elements that contain the node j. The local recovery gradient and the absolute recovery gradient error take the form

$$\begin{aligned} GU_j &= \frac{1}{|\omega|} \cdot \sum_{j\in\omega} |\Omega_j| \cdot \nabla U_j \\ \bar{\tilde{e}}_j &= |\nabla U_j - GU_j| = \left(1 - \frac{1}{|\omega|} \cdot \sum_{j\in\omega} |\Omega_j| \right) \cdot |\nabla U_j| \end{aligned} \tag{15}$$

Assuming that the recovery gradient is more accurate than the numerical gradient, it follow that the global error estimate obeys to

$$\eta_\Omega^{ZZ} \leqslant \kappa \cdot \eta_\Omega, \quad \text{with } \kappa < 1 \tag{16}$$

Defining the global recovery error $\|\bar{\tilde{e}}_\Omega\|_{L_2}$ norm from Eq. (15) as above in Eqs. (13) and (14), the estimate of the global true error η_Ω is bounded as

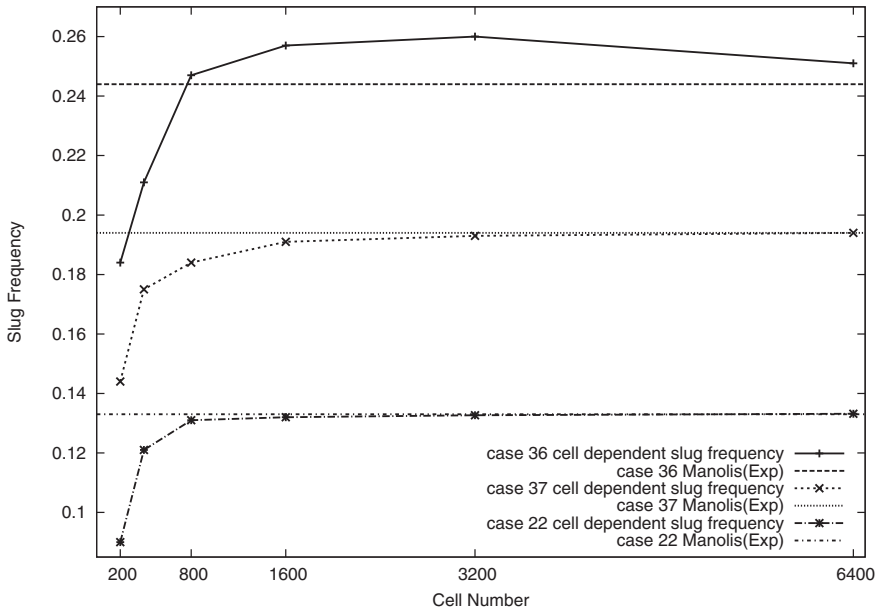


Fig. 12. Slug frequency as function of uniform meshes.

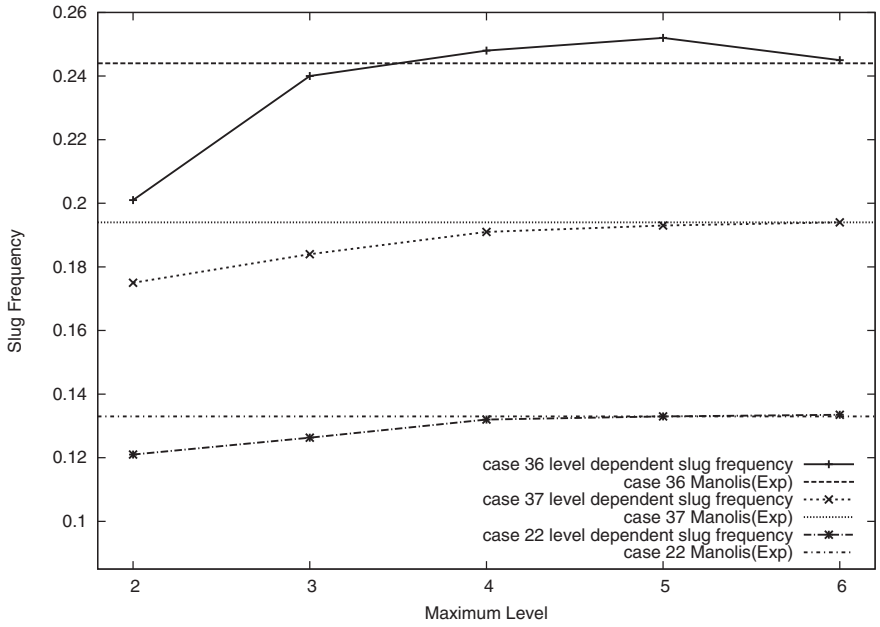


Fig. 13. Slug frequency as function of grid levels: *AMR*.

$$\frac{1}{(1+\kappa)}\cdot\|\bar{\bar{e}}_{\Omega}\|_{L_2}\leqslant\eta_{\Omega}\leqslant\frac{1}{(1-\kappa)}\cdot\|\bar{\bar{e}}_{\Omega}\|_{L_2} \tag{17}$$

Condition in (16) and herein the double side bounded inequality in (17) are conditioned by the regularity of the grid at each level; formally $\kappa=\kappa(\mathcal{G}_l)$.

The upper and lower bounds multiplicative parameters define the reliability $C_{rel}=1/(1+\kappa)$ and the efficiency $C_{eff}=1/(1-\kappa)$ of the ZZ error estimator.

4.1.3. An alternative error estimator

Instead of using *ZZ* error estimator, an alternative approach is adopted here. Concerning flows in long pipes with complex and bumpy geometries, building a robust parameter $\kappa(\Omega)$, therefore not recall, $\kappa(\Omega_l)$ that are problem independent is an unsolvable obstacle. In reality, topographies (domains) of extended systems vary from one problem to another.

In the present study, the grid refinement is subjected to a threshold on gradient truncation error in the following way; an average gradient norm L_2 is calculated for each half of each grid level l

$$\begin{aligned}\overline{\nabla U}^2_{l,-}&=2\cdot\sum_{i=2}^{nl/2}(U_i-U_{i-1})^2/nl\\ \overline{\nabla U}^2_{l,+}&=2\cdot\sum_{i=nl/2+1}^{nl}(U_i-U_{i-1})^2/nl\end{aligned} \tag{18}$$

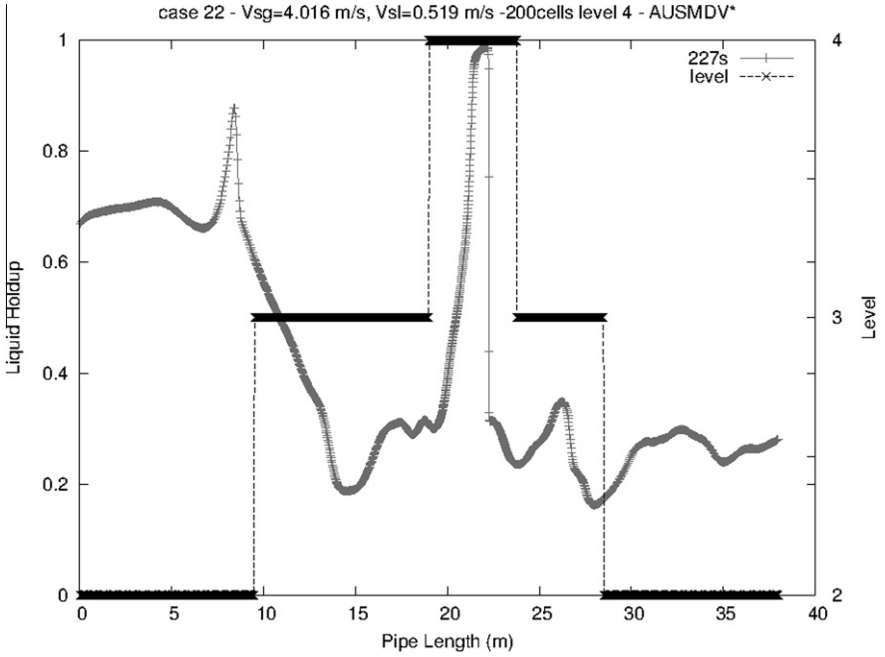


Fig. 14. Liquid distribution and grid level along the pipe at $t = 227$ s.

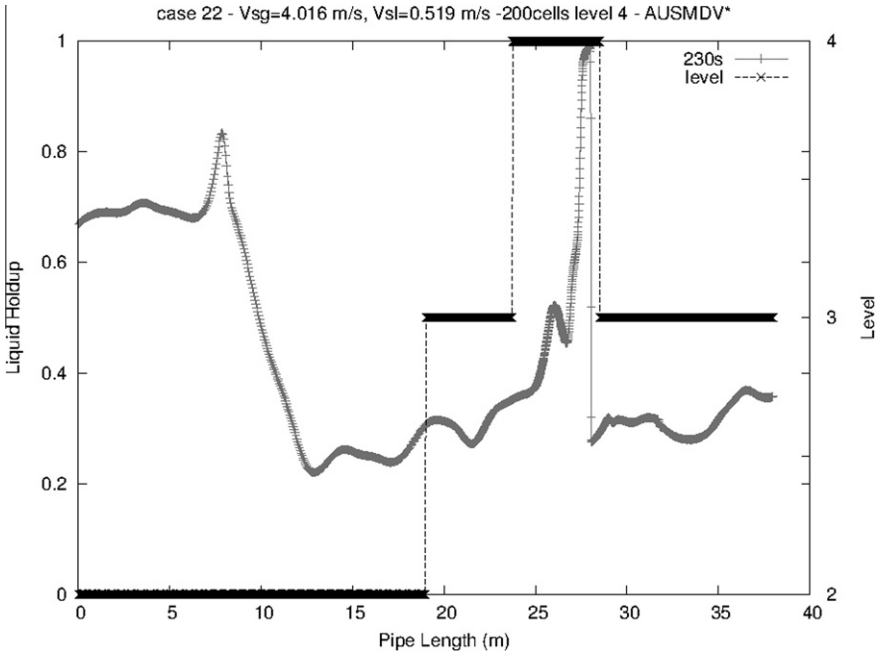


Fig. 15. Liquid distribution and grid level along the pipe at $t = 230$ s.

where nl is the number of cells in the grid level l. A half grid level l will be flagged for refinement if its average gradient $\overline{\nabla U}^2_{l,\pm}$ is greater or equal than to the following threshold:

$$\eta_l = \left(\sum_{k=1}^{l} \overline{\nabla U}^2_{k,+} + \overline{\nabla U}^2_{k,-} \right)^{\frac{1}{2}} \Big/ \sqrt{2N_l} \tag{19}$$

where N_l is the total number of cells which make up the level l. If the average gradient of half grid that is refined falls below the threshold in (19), the fine grid is destroyed. Eqs. (18) and (19) constitute the gradient criteria. However, as it will be shown below, this criteria on its own is unable to ensure an efficient treatment of slugs distribution and slugs frequency. To do so, it is supplied with a detector that rely on *Kelvin–Helmholtz KH* instability condition.

In series of investigations, Taitel and Dukler (1976), Barnea and Taitel (1992, 1993, 1994) amongst many others, studied transient flows in horizontal and inclined pipes. The authors performed stability analyses of interfacial waves in *two-fluid-model* that lead to understand and trace transition from stratified to slug flow regimes. *Kelvin–Helmholtz* stability criterion were obtained in both inviscid (IKH) and viscous (VKH) fluids, we refer to these by *Tai-*

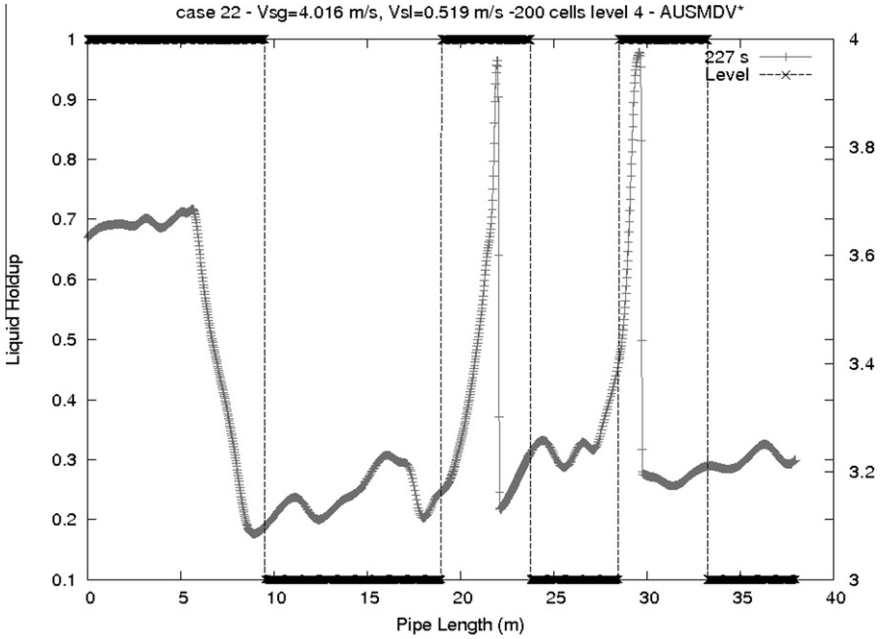


Fig. 16. Liquid distribution and grid level along the pipe at $t = 227$ s.

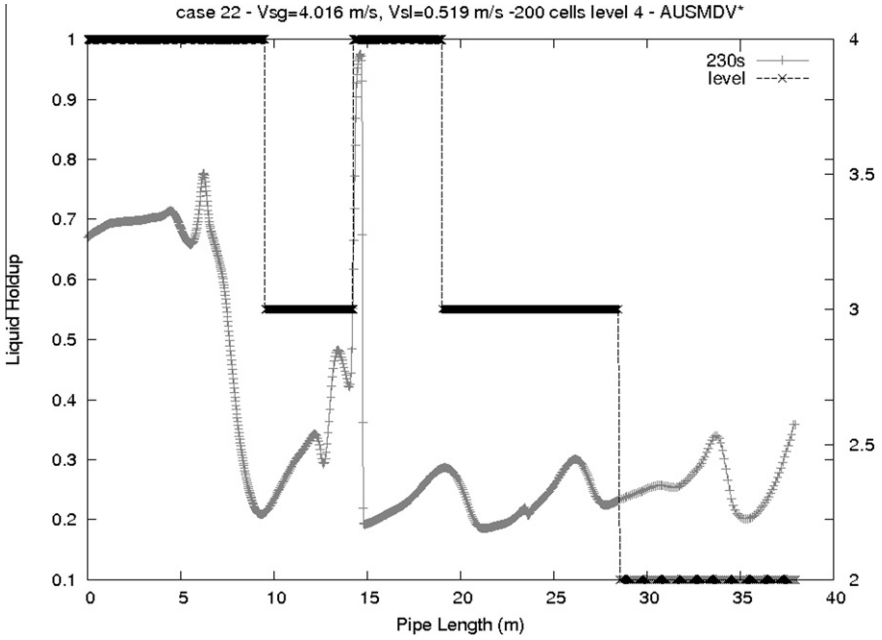


Fig. 17. Liquid distribution and grid level along the pipe at $t = 230$ s.

tel–Dukler–Kelvin–Helmholtz, *TD–KH* condition. These are based on the slip velocity and are a good indicator for the detection of the inception of slugs and hence for the location of the regions that require adaptive refinement.

In the present work, gas water two phase flow is considered and the indicator below for the gas velocity is adopted

$$u_g^2 \geqslant \left(1 - \frac{h_l}{D}\right)^2 \frac{(\rho_l - \rho_g) \cdot g \cos\beta}{\rho_g} \cdot \frac{\alpha_g}{d\alpha_l / dh_l} \tag{20}$$

where D is the pipe diameter and h_l stands for the liquid height in the pipe section.

In conjunction with the condition above, the gradient criteria in Eq. (18) with the inequality (19) ensures accurate treatment of slug fronts and tails as well as slug frequencies.

5. Numerical evaluations: Hydrodynamic slug cases

We performed numerical simulations with initial and boundary conditions corresponding to data obtained from experiments that were carried out by Manolis (1995). In these experiments, various flow rates of air/water in 38 m horizontal pipe with internal diameter $D = 0.078$ m at atmospheric pressure were analysed. Here,

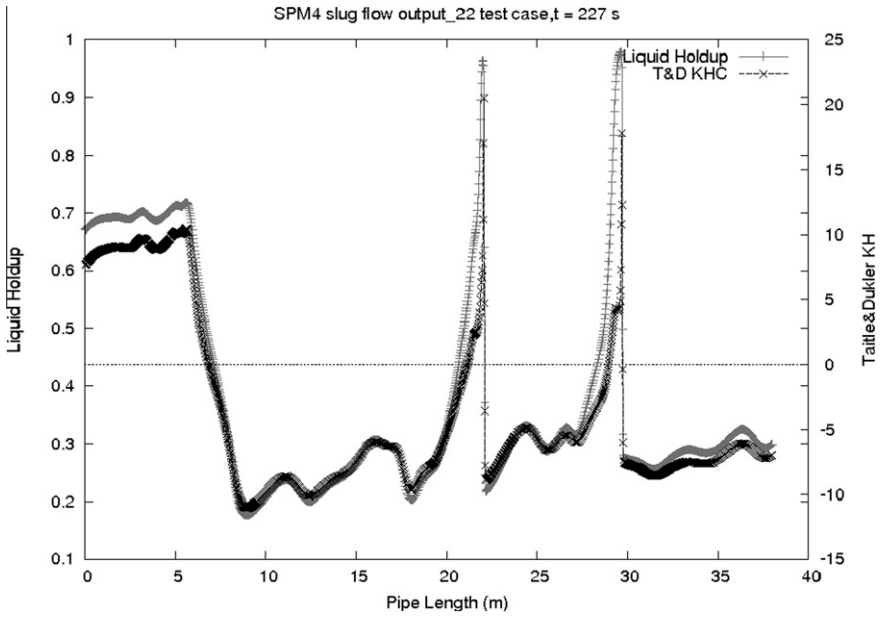


Fig. 18. Liquid distribution and *TD–KH* condition along the pipe at $t = 227$ s.

Table 3
Performance for slug flow *case*22: *TD–KH+* Gradient error criteria.

Level of refinement	Number of cells	Time		Speed-up	Slug frequency	
		Uniform	AMR		Uniform	AMR
1	200	00 h 07 min	–	–	0.09	–
2	400	00 h 28 min	00 h22 min	1.28	0.121	0.121
3	800	01 h 45 min	01 h6 min	1.6	0.131	0.1263
4	1600	06 h 57 min	03 h30 min	2.21	0.132	0.132
5	3200	15 h 23 min	04 h24 min	3.52	0.1327	0.133
6	6400	37 h 52 min	07 h29 min	5.08	0.1332	0.1335

Table 4
Performance for slug flow *case*36: *TD–KH+* Gradient error criteria.

Level of refinement	Number of cells	Time		Speed-up	Slug requency	
		Uniform	AMR		Uniform	AMR
1	200	00 h 05 min	–	–	0.184	–
2	400	00 h 24 min	00 h19 min	1.26	0.211	0.201
3	800	01 h 20 min	00 h53 min	1.52	0.247	0.240
4	1600	05 h 18 min	02 h28 min	2.14	0.257	0.248
5	3200	12 h 44 min	03 h40 min	3.40	0.260	0.252
6	6400	29 h 17 min	06 h00 min	4.91	0.251	0.245

Table 5
Performance for slug flow *case*37:*TD–KH+* Gradient error criteria.

Level of refinement	Number of cells	Time		Speed-up	Slug requency	
		Uniform	AMR		Uniform	AMR
1	200	00 h 07 min	–	–	0.144	–
2	400	00 h 29 min	00 h22 min	1.31	0.175	0.170
3	800	01 h 34 min	00 h55 min	1.69	0.184	0.175
4	1600	06 h 15 min	02 h30 min	2.35	0.191	0.19
5	3200	19 h 10 min	05 h00 min	3.81	0.193	0.192
6	6400	44 h 22 min	08 h16 min	5.45	0.194	0.193

three test cases referenced in tables below were selected for validations; Pipe geometry and fluids properties are shown in Table 1.

Initial conditions for three cases are specified in Table 2. It depicts also water volume fractions, these α_l correspond to the numerical input.

Barnea (1991) described the interface response and its behaviour under the effect of the viscous and inviscid instabilities. The author was able to distinguish bounded regions of instability in terms of superficial gas and liquid velocity, these are shown in Graph (Fig. 5).

Fig. 5 reproduced below is adapted to the actual pipe geometry, the blue dots reported on the graph re er to the three cases under study and show that these are in the slug flow regime zone.

5.1. Effect of gradient error estimator on refinement

To assess the effectiveness of the refinement error criteria in use; it is necessary to provide a practical evaluation. Hence, the spectra of the liquid hold up during over period of time were plotted. The liquid volume fraction is monitored by numerical sensor located at $x = 29$ m downstream and the role of the application of the complete or partial error criterion are compared below.

5.1.1. Gradient error estimator effect: Case 22

Fig. 6 shows the spectrum obtained with uniform mesh of 1600 cells.

Fig. 7 shows the spectrum of the liquid holdup along the pipe when *AMR* with level $l = 4$ is applied along only with the gradient error criteria Eqs. (18) and (19).

It is noticeable from the spectrum Fig. 7, that the gradient criteria alone is unable to locate regions with steep gradients.

The gradient error criteria has tendency to render the spectrum quasi-periodic by smearing out a non-negligible number of slugs and waves.

Fig. 8 depicts the same spectrum when the error gradient in conjunction with *Kelvin–Helmholtz* is applied. The application of the complete criteria along with *AMR* shows no major effect on the spectrum distribution as global, only local changes can be observed.

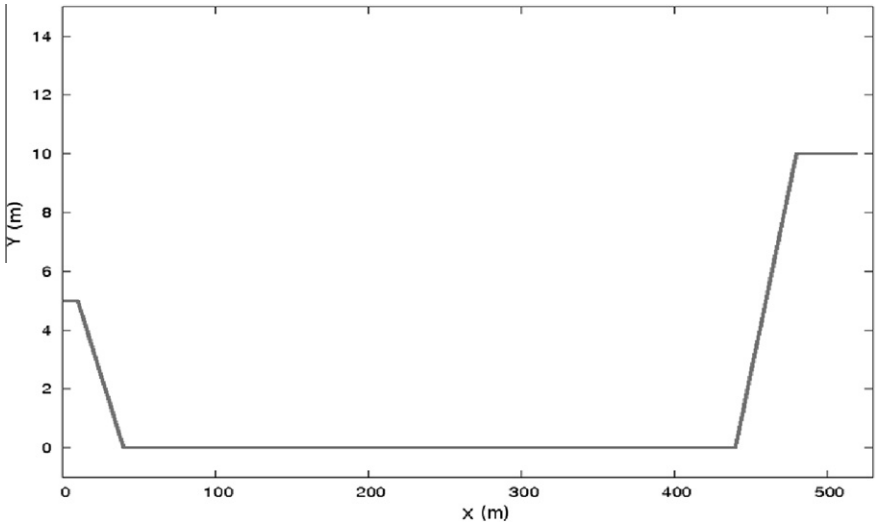


Fig. 19. Topography of the pipeline $L = 520$ (m).

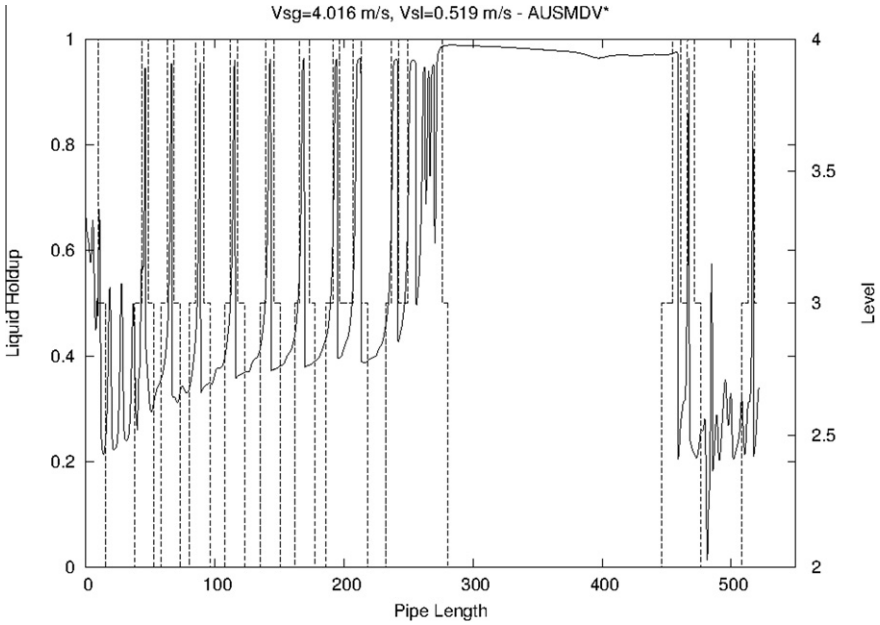


Fig. 20. Liquid distribution and grid level along the pipeline at $t = 52$ s.

5.1.2. Gradient error estimator effect: Case 36

Fig. 9, below illustrates the trace of the liquid hold up obtained with uniform mesh under the flow conditions of the case 36.

Fig. 10 portrays the effect of gradient error criteria while Fig. 11 shows the liquid hold up when the complete error criteria when *AMR* is applied. The final grid level is $l = 4$ and the initial mesh has 200 cells.

As in the previous case, the use gradient error criteria alone results in smearing out several slugs and waves, leading to erroneous values of slug characteristics.

In this case the spectrum in Fig. 9 is more regular than it is for the *Case*22, Fig. 10 shows the obvious filtering effect of gradient error criteria on slug frequency and on the baseline of the liquid holdup; globally, one in three slugs was smeared out.

Combining gradient error criterion with *K–H* condition is necessary for accurate estimation of slug frequencies. Comparing Fig. 8 with Figs. 6 and 11 with Fig. 9 shows that applying the present *AMR* technique alters the liquid holdup spectra only locally but globally it preseves their mean features.

5.2. Slug frequency

A series of computations were completed in order to evaluate slug frequencies dependency on mesh size. This was tested in uniform meshes as well as with adaptive mesh refinement. In Figs. 12 and 13, horizontal dashed lines represent experimental measurement of slug frequencies obtained by Manolis (1995).

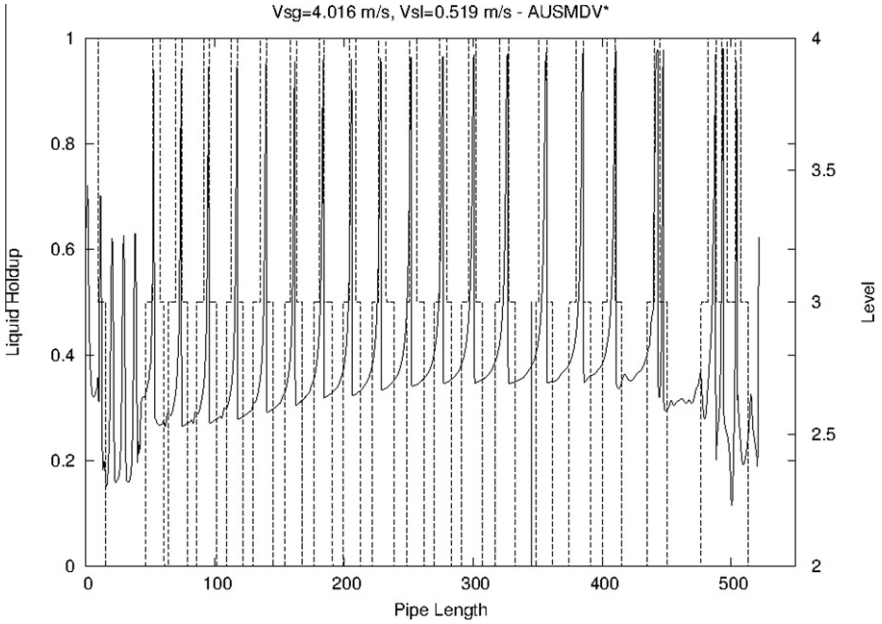


Fig. 21. Liquid distribution and grid level along the pipeline at $t = 102$ s.

Table 6
Performance for slug flow *case*22 in non-uniform pipe.

Level of refinement	Number of cells	Time		Speed-up
		Uniform	AMR	
1	1000	05 h 13 min	–	–
2	2000	20 h 48 min	15 h25 min	1.35
3	4000	68 h 59 min	29 h55 min	2.34
4	8000	153 h 43 min	41 h34 min	3.68
5	16000	376 h 35 min	73 h12 min	5.15
6	32000	917 h 45 min	124 h52 min	7.39

Fig. 12 shows the evolution of the slug frequency for five uniform meshes. The agreement with the data measurement is acceptable except for the *Case*36, where numerical frequency is overestimated and it seems that higher uniform meshes are required in order to reach a satisfactory agreement with data.

Fig. 13 shows the behaviour of the slug frequency for five grid levels obtained with *AMR* technique. The agreement with the data measurement is acceptable even for the case referenced 36, where numerical frequency is overestimated first and then converges to the measured value.

Both figures show that slug frequency is mesh size dependent, in all cases the numerical frequencies converge to the values of observed frequencies.

5.3. Taitel–Dukler–Kelvin–Helmholtz condition effect on refinement

In this subsection, we attempt to comprehend the effect of *TD–KH* condition on the griding. Figs. 14 and 15 show three sequence the liquid distribution and the grid levels along the pipe when only the gradient error criteria is applied.

The following two sequences show the liquid holdup distribution and the grid levels along the pipe when the gradient error criteria and *TD–KH* condition are applied.

Comparing Figs. 14 and 15 with Figs. 16 and 17 we actually see that in the latter figures, the inlet part of the pipe uses always high grid level while in the former a low grid level used. In using gradient error criteria alone leads to refining areas that are not required to, resulting in CPU time consuming, this is clearly shown in Figs. 14 and 15, where a large parts of the pipe between successive slugs are refined. Applying the complete criteria, (*TD–KH* condition + gradient error criteria) allows for tracing slugs and use high grid level in their vicinities.

Liquid holdup distributions and corresponding *TD–KH* condition are depicted in the graphs below, the horizontal line correspond to the marginal stability *TD–KH* condition.

Fig. 18 shows that *TD–KH* condition is an appropriate marker for slugs detection, hence a good indicator for mesh refinement. The marginal stability line (horizontal line) indicates the border separating *Stratified*/*Slug* flow transition. Zones below this line correspond to stratified regions while situations above it are unstable zones and are potential slug regions, therefore candidates for refinement.

6. Results and performances synopsis

In this section, a summary of the results obtained in this work are reported and compared to the data and the performances gained with *AMR* technique.

Simulations were carried out for flow time $T \approx 250$ s, Tables 3–5 display the execution time for the integration of slug flows cases 22, 36 and 37.

These tables show that the computed frequencies are in good agreement with the experimental measurements, more significantly, the performances gained in the computational times were about $\approx$5 times faster compared to computational times needed when uniform meshes were used.

The *case*36 simulation presents a characteristic were the calculated frequencies are overestimated compared to experimental values, this was observed with uniform mesh as well as with *AMR*.

Performances in the *Speed-up* for these flow cases are of the same order, that was expected regarding that flows characteristics are comparable.

7. Slug flow in non-uniform pipe

In this section, we consider a slug flow with conditions of the *case*22 flow in a non-uniform pipe with length of $l \approx 500$ m. The topology of the system considered is illustrated in Fig. 19 below.

The aim here is to assess the capability and sensitivity of *TD–KH* with gradient error criteria along with the *AMR* technique to detect the inception of topological slugs, or slugs resulting from accidental terrains. To do so snapshots of the liquid distribution and grid level behaviour in the pipeline are presented at different instants in Figs. 20 and 21.

Graphs (Figs. 20 and 21) show two stages of the time evolution of the liquid holdup with adaptive grid level along the system. Particularly in Fig. 20 and in Fig. 21, the onset of slugging occurs at the start of the horizontal part just after the downward and in the vicinity of the upward regions, these areas are primary candidates for refinement.

As the simulation proceeds, the number of flagged regions for refinement increases resulting in the CPU consumption. It is recommended to limit the maximum levels required in order to keep a good trade-off between the *Speed-up*, accuracy and efficiency.

The comparison of performances between simulations using uniform mesh with using *AMR* technique are illustrated in Table 6.

8. Conclusion

We have applied an adaptive mesh refinement technique *AMR* to two phase flow '*two-fluids-model*' combined with *AUSMDV** numerical integrator.

The description includes a global error-estimator that is double side bounded. The recovery approach is not convenient in extended and non uniform pipelines.

The double sided condition in (17) is very appealing from mathematical point of view but the irregularity of the topology makes the *mesh–regularity* condition associated to it an unrealistic condition. Furthermore, the condition in (17) is *topology–dependent*, and this is the least we wanted. Since we are dealing with slug flow regimes, rather, we used a one side bounded gradient norm error in conjunction with a physical *TD–KH* stability condition. This combination proved to be efficient and its results showed to be in good agreement with data.

Furthermore, the implementation of *TD–KH* condition require less effort than implementing any *recovry-inperpolation* method, moreover, it has the advantage of been a *local condition* and is an *a priori* indicator for refinement.

We showed that global gradient norm error estimator is unable to detect the main features of the flows that are characterised by discontinuities or steep gradients. Concerning flow in extended and non-uniform pipe, the approach showed a remarkable ability to detect slugs resulting from the terrain.

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